

Lab: Filtering, Sorting, and Grouping Data in Pandas for Effective Analysis of Real-World Datasets (Superstore Dataset)

Problem Statement:

In real-world datasets, data analysis requires filtering, sorting, and grouping to extract meaningful insights. This lab will explore these operations using Pandas on the Superstore dataset to gain useful business insights.

Objective:

- Learn how to filter data based on specific conditions.
- Sort data to identify trends and patterns.
- Group data for aggregated analysis.
- Analyze sales performance based on different factors.

Requirements:

- Python installed (version 3.x recommended).
- Pandas and NumPy libraries installed (`pip install pandas numpy`).
- Jupyter Notebook (optional but recommended for running the lab).
- Superstore dataset (CSV file) downloaded.

Part 1: Importing Libraries and Loading Data

```
In [ ]: # import necessary Library
import pandas as pd
import numpy as np
```

```
In [6]: # Load the dataset
df = pd.read_csv('Dataset/SampleSuperstore.csv')
print("Dataset Loaded Successfully")
df.head()
```

Dataset Loaded Successfully

Out[6]:

	Ship Mode	Segment	Country	City	State	Postal Code	Region	Category	Category Name
0	Second Class	Consumer	United States	Henderson	Kentucky	42420	South	Furniture	Bookcase
1	Second Class	Consumer	United States	Henderson	Kentucky	42420	South	Furniture	Chair
2	Second Class	Corporate	United States	Los Angeles	California	90036	West	Office Supplies	Laptop
3	Standard Class	Consumer	United States	Fort Lauderdale	Florida	33311	South	Furniture	Table
4	Standard Class	Consumer	United States	Fort Lauderdale	Florida	33311	South	Office Supplies	Stapler

Part 2: Filtering Data

In [8]:

```
# Filter for orders with Sales greater than $500
print("Orders with Sales greater than $500:")
high_sales = df[df['Sales'] > 500]
high_sales.head()
```

Orders with Sales greater than \$500:

Out[8]:

	Ship Mode	Segment	Country	City	State	Postal Code	Region	Category	Category Name
1	Second Class	Consumer	United States	Henderson	Kentucky	42420	South	Furniture	Bookcase
3	Standard Class	Consumer	United States	Fort Lauderdale	Florida	33311	South	Furniture	Table
7	Standard Class	Consumer	United States	Los Angeles	California	90032	West	Technology	Laptop
10	Standard Class	Consumer	United States	Los Angeles	California	90032	West	Furniture	Table
11	Standard Class	Consumer	United States	Los Angeles	California	90032	West	Technology	Laptop

In [9]:

```
# Filter for orders from a specific region (e.g., 'West')
print("Orders from the West region:")
west_orders = df[df['Region'] == 'West']
west_orders.head()
```

Orders from the West region:

Out[9]:

	Ship Mode	Segment	Country	City	State	Postal Code	Region	Category	Sales
2	Second Class	Corporate	United States	Los Angeles	California	90036	West	Office Supplies	La
5	Standard Class	Consumer	United States	Los Angeles	California	90032	West	Furniture	Furnish
6	Standard Class	Consumer	United States	Los Angeles	California	90032	West	Office Supplies	
7	Standard Class	Consumer	United States	Los Angeles	California	90032	West	Technology	Pho
8	Standard Class	Consumer	United States	Los Angeles	California	90032	West	Office Supplies	Bin



Part 3: Sorting Data

```
In [10]: # Sorting by Sales in descending order
print("Top 5 highest Sales orders:")
sorted_sales = df.sort_values(by='Sales', ascending=False)
sorted_sales.head()
```

Top 5 highest Sales orders:

Out[10]:

	Ship Mode	Segment	Country	City	State	Postal Code	Region	Category
2697	Standard Class	Home Office	United States	Jacksonville	Florida	32216	South	Technology
6826	Standard Class	Corporate	United States	Lafayette	Indiana	47905	Central	Technology
8153	First Class	Consumer	United States	Seattle	Washington	98115	West	Technology
2623	First Class	Home Office	United States	New York City	New York	10024	East	Technology
4190	Standard Class	Consumer	United States	Newark	Delaware	19711	East	Technology



```
In [11]: # Sorting by Profit in ascending order
print("Bottom 5 least profitable orders:")
sorted_profit = df.sort_values(by='Profit', ascending=True)
sorted_profit.head()
```

Bottom 5 least profitable orders:

Out[11]:

	Ship Mode	Segment	Country	City	State	Postal Code	Region	Category	Count
7772	Standard Class	Consumer	United States	Lancaster	Ohio	43130	East	Technology	1
683	Same Day	Corporate	United States	Burlington	North Carolina	27217	South	Technology	1
9774	Standard Class	Consumer	United States	San Antonio	Texas	78207	Central	Office Supplies	1
3011	Standard Class	Home Office	United States	Louisville	Colorado	80027	West	Technology	1
4991	Standard Class	Corporate	United States	Chicago	Illinois	60653	Central	Office Supplies	1



Part 4: Grouping Data

```
In [14]: # Group by Category and calculate total sales
print("Total Sales per Category:")
category_sales = df.groupby('Category')['Sales'].sum()
category_sales
```

Total Sales per Category:

```
Out[14]: Category
Furniture      741999.7953
Office Supplies 719047.0320
Technology     836154.0330
Name: Sales, dtype: float64
```

```
In [16]: # Group by State and find the average profit
print("Average Profit per State:")
state_profit = df.groupby('State')['Profit'].mean()
state_profit
```

Average Profit per State:

```
Out[16]: State
Alabama      94.865989
Arizona      -15.303235
Arkansas      66.811452
California    38.171608
Colorado     -35.867351
Connecticut   42.823071
Delaware     103.930988
District of Columbia 105.958930
Florida      -8.875461
Georgia       88.315453
Idaho         39.367767
Illinois     -25.625787
Indiana       123.375411
Iowa          39.460397
Kansas        34.851813
Kentucky      80.573357
Louisiana     52.288150
Maine         56.810775
Maryland      66.963608
Massachusetts 50.262975
Michigan      95.934069
Minnesota     121.608847
Mississippi   59.867475
Missouri      97.518341
Montana       122.221900
Nebraska      53.607742
Nevada        85.045279
New Hampshire 63.203807
New Jersey    75.176260
New Mexico    31.273408
New York      65.637011
North Carolina -30.083985
North Dakota  32.878529
Ohio          -36.186304
Oklahoma      73.544788
Oregon        -9.600569
Pennsylvania  -26.507598
Rhode Island  130.100523
South Carolina 42.120395
South Dakota  32.902358
Tennessee     -29.189583
Texas         -26.121174
Utah          48.047802
Vermont       204.088936
Virginia      83.026564
Washington    66.013146
West Virginia 46.480400
Wisconsin      76.380004
Wyoming       100.196000
Name: Profit, dtype: float64
```

Part 5: Aggregating Data

```
In [17]: # Finding total and average sales per region
print("Total and Average Sales per Region:")
region_sales = df.groupby('Region')['Sales'].agg(['sum', 'mean'])
print(region_sales)
```

Total and Average Sales per Region:

	sum	mean
Region		
Central	501239.8908	215.772661
East	678781.2400	238.336110
South	391721.9050	241.803645
West	725457.8245	226.493233